ADEROUTE BILLING SYSTEM - MIKROTIK INSTALLATION GUIDE

Complete Step-by-Step Documentation

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1. PREREQUISITES

Hardware Requirements

- MikroTik Router (RB951, hAP, or any model with WiFi)
- Internet connection (Fiber/DSL/LTE) with ethernet cable
- Computer with WinBox or SSH access

Software Requirements

- RouterOS v6.40+ or v7.x
- Your Aderoute billing system URL (e.g., https://yourdomain.vercel.app)
- API access enabled on MikroTik

Network Plan

Component	IP Address	Purpose
Router LAN/WiFi	192.168.23.1/24	Hotspot network
DHCP Pool	192.168.23.2-192.168.23.254	Customer IPs
WAN Port (ether1)	DHCP from ISP	Internet source

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2. ROUTER INITIAL SETUP

Step 2.1: Connect to Router

1. Connect ethernet cable from PC to ether2 (LAN port)
2. Open WinBox, click **Neighbors** tab
3. Click on router's MAC address and connect
4. Username: admin, Password: (blank)

Step 2.2: Reset to Factory Defaults (Optional but Recommended)

text

```
/system reset-configuration skip-backup=yes no-defaults=no
```

Wait for router to reboot (2 minutes), then reconnect.

Step 2.3: Set Router Identity

text

```
/system identity set name="Aderoute-Hotspot"
```

Step 2.4: Configure WAN Interface (Internet)

text

```
# Add DHCP client on ether1 (WAN port)
/ip dhcp-client add interface=ether1 disabled=no
```

Verify IP is obtained

/ip address print

Expected output shows 192.168.x.x/24 on ether1 (or your ISP's IP range).

Step 2.5: Configure LAN Bridge and IP

text

```
# Create bridge for hotspot network
/interface bridge add name=bridge-hotspot

# Assign IP to bridge
/ip address add address=192.168.23.1/24 interface=bridge-hotspot

# Add LAN ports to bridge
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=ether5
```

Step 2.6: Add DHCP Server for Hotspot Network

text

```
# Create IP pool
/ip pool add name=hotspot-pool ranges=192.168.23.2-192.168.23.254

# Add DHCP server
/ip dhcp-server add name=dhcp-hotspot interface=bridge-hotspot address-pool=hotspot-pool

# Set DHCP lease time (10 minutes)
/ip dhcp-server set dhcp-hotspot lease-time=10m
```

Enable DHCP server

/ip dhcp-server enable dhcp-hotspot

Step 2.7: Configure NAT for Internet Sharing

text

Masquerade traffic going out to WAN

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

Basic DNS settings

/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes

---

3. HOTSPOT CONFIGURATION

Step 3.1: Enable API Service (For Billing System)

text

Enable API on default port 8728

/ip service enable api

/ip service set api port=8728

Verify

/ip service print where name=api

Step 3.2: Create Hotspot Profile

text

Add server profile

/ip hotspot profile add name=hsprof1 hotspot-address=192.168.23.1 dns-name=hotspot.local

Step 3.3: Create Hotspot Server

text

Add hotspot server on bridge-hotspot

/ip hotspot add name=hs1 interface=bridge-hotspot address-pool=hotspot-pool profile=hsprof1

Enable hotspot

/ip hotspot enable hs1

Step 3.4: Create User Profiles (Plans)

text

Create time-based plans with speed limits

/ip hotspot user profile add name="1HourPass" rate-limit="5M/5M" idle-timeout=1h

/ip hotspot user profile add name="3HoursPass" rate-limit="5M/5M" idle-timeout=3h

/ip hotspot user profile add name="6HoursPass" rate-limit="10M/10M" idle-timeout=6h

/ip hotspot user profile add name="24Hours" rate-limit="10M/10M" idle-timeout=24h

/ip hotspot user profile add name="WeeklyPass" rate-limit="15M/15M" idle-timeout=168h

/ip hotspot user profile add name="MonthlyPass" rate-limit="20M/20M" idle-timeout=720h

Verify all profiles

/ip hotspot user profile print

Step 3.5: Create API User for Billing System

text

Create dedicated user group for API access

/user group add name=billing-api policy=read,write,api

Create API user

/user add name=aderoute password=your_secure_password group=billing-api

Restrict API access to billing server IP (optional but recommended)

/ip service set api address=192.168.23.0/24

---

4. WIFI SETUP

Step 4.1: Configure Wireless Interface

text

Enable wireless interface

/interface wireless enable wlan1

Set to access point mode with open network (no password)

/interface wireless set wlan1 mode=ap-bridge ssid="Aderoute" band=2ghz-b/g/n disabled=no

Create open security profile

/interface wireless security-profiles add name=open mode=none

Apply open profile

/interface wireless set wlan1 security-profile=open

Step 4.2: Add WiFi to Hotspot Bridge

text

Add wlan1 to the hotspot bridge

/interface bridge port add bridge=bridge-hotspot interface=wlan1

Verify

/interface bridge port print

Step 4.3: Optional - Create Virtual WiFi for Management

text

Hide default SSID if you want only "Aderoute" visible

```
/interface wireless set wlan1 hide-ssid=yes

# Create visible hotspot WiFi
/interface wireless add name=wlan-hotspot master-interface=wlan1 mode=ap-bridge ssid="Aderoute" security-profile=open disabled=no

# Add to bridge
/interface bridge port add bridge=bridge-hotspot interface=wlan-hotspot
```

---

5. BILLING SYSTEM INTEGRATION

Step 5.1: Configure Walled Garden (Allow Billing Site Without Login)

text

```
# Add your billing domain to bypass hotspot login
/ip hotspot walled-garden ip add dst-host=yourdomain.vercel.app
/ip hotspot walled-garden ip add dst-host=*.yourdomain.vercel.app

# Add any backend services your billing system uses
/ip hotspot walled-garden ip add dst-host=convex.cloud
/ip hotspot walled-garden ip add dst-host=*.convex.cloud
/ip hotspot walled-garden ip add dst-host=vercel.app
/ip hotspot walled-garden ip add dst-host=*.vercel.app

# Add common CDN and API services
/ip hotspot walled-garden ip add dst-host=cloudflare.com
/ip hotspot walled-garden ip add dst-host=*.cloudflare.com
/ip hotspot walled-garden ip add dst-host=googleapis.com
```

/ip hotspot walled-garden ip add dst-host=*.googleapis.com

Step 5.2: Create Custom Redirect Page (Optional - Redirect to Billing System)

text

Create HTML redirect file

/file set hotspot/login.html content="<html><head><meta http-equiv='refresh' content='0; url=https://yourdomain.vercel.app'></head><body>Redirecting to Aderoute...</body></html>"

Apply custom page

/ip hotspot profile set hsprof1 html-directory=hotspot

Reset hotspot to apply changes

/ip hotspot reset-html

Step 5.3: Configure DNS for Hotspot (Optional)

text

Create simple local DNS entry

/ip dns static add name=hotspot.aderoute.com address=192.168.23.1

---

6. ADVANCED CONFIGURATIONS

6.1 Bandwidth Management (Per User Limits)

Already set in profiles. To modify:

text

Example: Change 1HourPass to 10Mbps

/ip hotspot user profile set 1HourPass rate-limit="10M/10M"

6.2 Idle Timeout Settings

text

Set global idle timeout (disconnect inactive users)

/ip hotspot set hs1 idle-timeout=15m

Keepalive timeout (ensure connection stays alive)

/ip hotspot set hs1 keepalive-timeout=2m

6.3 Multiple SSIDs (Premium vs Regular)

text

Create premium WiFi with faster speeds

/interface wireless add name=wlan-premium master-interface=wlan1 mode=ap-bridge ssid="Aderoute-Premium" security-profile=open

Add to separate hotspot or same with different profile

/interface bridge port add bridge=bridge-hotspot interface=wlan-premium

Create premium user profiles with higher speeds

/ip hotspot user profile add name="Premium-1Hour" rate-limit="50M/50M" idle-timeout=1h

6.4 Backup and Restore Configuration

text

Backup full configuration

/system backup save name=aderoute-config.backup

Export as script (readable format)

/export file=aderoute-setup

Restore from backup

/system backup load name=aderoute-config.backup

---

7. TESTING & VERIFICATION

7.1 Verify Hotspot Status

text

Check hotspot server is running

/ip hotspot print

Expected: hs1 should NOT have X (disabled) flag.

7.2 Verify WiFi is Broadcasting

text

Check wireless interfaces

/interface wireless print

Expected: wlan1 or wlan-hotspot shows R (running).

7.3 Test with Mobile Phone

1. Scan for WiFi networks
2. Find **"Aderoute"** (or your SSID name)
3. Connect (no password required)
4. Open browser - should redirect to your billing system
5. Billing system should load fully

7.4 Test User Creation via API

text

Test API connection (from billing server)

/tool fetch url=https://192.168.23.1:8728/rest/system/resource port=8728 user=aderoute password=your_secure_password

Create test user manually on router

/ip hotspot user add name=test001 password=pass123 profile=1HourPass

Verify user created

/ip hotspot user print

7.5 Test Customer Login

1. Connect phone to "Aderoute" WiFi
2. Go to billing site, purchase plan
3. Use provided credentials to login at http://192.168.23.1
4. Internet should work

---

8. TROUBLESHOOTING

Issue: WiFi not visible

text

Enable wireless interface

/interface wireless enable wlan1

Check SSID is not hidden

/interface wireless print where name=wlan1

Look for "hide-ssid=no"

Issue: No redirect to billing system

text

Check walled garden entries

/ip hotspot walled-garden print

Verify custom HTML is set

/ip hotspot profile print

Reset hotspot

/ip hotspot reset-html

Issue: Billing system loads but WebSocket fails

text

Add missing WebSocket/backend domains

/ip hotspot walled-garden ip add dst-host=wss.yourdomain.com

/ip hotspot walled-garden ip add dst-host=*.convex.cloud

Check DNS resolution

/ping yourdomain.com

Issue: Customers get "No Internet" after login

text

Check NAT rule

/ip firewall nat print where chain=srcnat

Verify WAN has internet

/ping 8.8.8.8

Check DNS

/ip dns print

Issue: API connection fails

text

Verify API service is enabled

/ip service print where name=api

Check API user permissions

/user print where name=aderoute

Test API manually

/tool fetch url=http://192.168.23.1:8728/rest/system/resource user=aderoute password=your_password

Issue: Hotspot stops after reboot

text

Ensure all services are enabled

/ip hotspot enable hs1

/ip dhcp-server enable dhcp-hotspot

/interface wireless enable wlan1

Save configuration

/system backup save name=working-config

---

9. QUICK DEPLOYMENT SCRIPT

Copy and paste this entire block to deploy a new MikroTik router:

text

=== ADEROUTE QUICK DEPLOYMENT SCRIPT ===

Replace "yourdomain.vercel.app" with your actual billing URL

Basic setup

/system identity set name="Aderoute-Hotspot"

/ip dhcp-client add interface=ether1 disabled=no

Bridge and IP

/interface bridge add name=bridge-hotspot

```
/ip address add address=192.168.23.1/24 interface=bridge-hotspot

# Add ports to bridge
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=ether5

# DHCP
/ip pool add name=hotspot-pool ranges=192.168.23.2-192.168.23.254
/ip dhcp-server add name=dhcp-hotspot interface=bridge-hotspot address-pool=hotspot-pool
/ip dhcp-server set dhcp-hotspot lease-time=10m
/ip dhcp-server enable dhcp-hotspot

# NAT and DNS
/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes

# API
/ip service enable api
/ip service set api port=8728
/user group add name=billing-api policy=read,write,api
/user add name=aderoute password=ChangeMe123 group=billing-api

# Hotspot
/ip hotspot profile add name=hsprof1 hotspot-address=192.168.23.1
```

```
/ip hotspot add name=hs1 interface=bridge-hotspot address-pool=hotspot-pool profile=hsprof1
/ip hotspot enable hs1

# User profiles (plans)
/ip hotspot user profile add name="1HourPass" rate-limit="5M/5M" idle-timeout=1h
/ip hotspot user profile add name="3HoursPass" rate-limit="5M/5M" idle-timeout=3h
/ip hotspot user profile add name="6HoursPass" rate-limit="10M/10M" idle-timeout=6h
/ip hotspot user profile add name="24Hours" rate-limit="10M/10M" idle-timeout=24h
/ip hotspot user profile add name="WeeklyPass" rate-limit="15M/15M" idle-timeout=168h
/ip hotspot user profile add name="MonthlyPass" rate-limit="20M/20M" idle-timeout=720h

# WiFi
/interface wireless enable wlan1
/interface wireless security-profiles add name=open mode=none
/interface wireless set wlan1 mode=ap-bridge ssid="Aderoute" security-profile=open
disabled=no
/interface bridge port add bridge=bridge-hotspot interface=wlan1

# Walled garden (UPDATE THIS WITH YOUR URL)
/ip hotspot walled-garden ip add dst-host=yourdomain.vercel.app
/ip hotspot walled-garden ip add dst-host=*.yourdomain.vercel.app
/ip hotspot walled-garden ip add dst-host=convex.cloud
/ip hotspot walled-garden ip add dst-host=*.convex.cloud
/ip hotspot walled-garden ip add dst-host=vercel.app

# Optional: Custom redirect page
```

```
/file set hotspot/login.html content="<html><head><meta http-equiv='refresh' content='0; url=https://yourdomain.vercel.app'></head><body>Redirecting...</body></html>"

/ip hotspot profile set hsprof1 html-directory=hotspot

/ip hotspot reset-html


# Save configuration

/system backup save name=aderoute-final


# === DEPLOYMENT COMPLETE ===
```

---

10. BILLING SYSTEM API INTEGRATION REFERENCE

API Endpoint (MikroTik Router)

text

URL: http://192.168.23.1:8728/rest

Authentication: Basic Auth (username: aderoute, password: your_secure_password)

Common API Commands for Your Billing System

Create Hotspot User:

text

```
POST /rest/ip/hotspot/user

{

  "name": "customer001",

  "password": "voucher123",

  "profile": "1HourPass"

}
```

List Active Users:

text

GET /rest/ip/hotspot/active

Delete Expired User:

text

DELETE /rest/ip/hotspot/user/customer001

Get User Status:

text

GET /rest/ip/hotspot/user/customer001

---

11. SECURITY RECOMMENDATIONS

1. **Change default admin password immediately:**

text

/user set admin password=YourStrongPassword

2. **Restrict API access to your billing server only:**

text

/ip service set api address=YOUR_BILLING_SERVER_IP/32

3. **Enable firewall to block unauthorized access:**

text

/ip firewall filter add chain=input protocol=tcp dst-port=8291 action=drop

/ip firewall filter add chain=input src-address=192.168.23.0/24 action=accept

4. **Regular backups:**

text

/system backup save name=backup-$(/system clock get date)

---

12. SUPPORT & CONTACT

For issues with:

- **MikroTik configuration:** Check MikroTik Wiki or forum

- **Aderoute billing system:** Your development team
- **API integration:** Review the API reference above

DOCUMENT VERSION

- **Version:** 1.0
- **Date:** April 2026
- **RouterOS Compatibility:** 6.40 - 7.x
- **Billing System:** Aderoute

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Congratulations! You now have complete documentation for deploying Aderoute on any MikroTik router.

End of Documentation

-check next public isp app url env variable for .env.local files of mpesa payment app and isp billing system app, -check next public isp app url env variable for convex dashboards of both

mpesa app and isp billing app -check next public isp app url env variable for vercel for both apps, -check the file for D:\Documents\CODE_WITH_ERICADESHH\GitHub\isp-billing-system\src\app\quick-pay\page.tsx, ensure the process.env.NEXT_PUBLIC_ISP_APP_URL is matching,